KIO Router AI — Project Progress Report

CUSTOMER MESSAGE ROUTING | WHOLE SYSTEM | PRESENTATION: MONDAY, 26 OCTOBER 2026

1. Project Objective and Scope

KIO Router AI routes each incoming customer message to the appropriate specialist department:

- **Support (Flag 0):** Problems with existing services, complaints, billing questions, account help.
- **Sales (Flag 1):** New purchases, upgrades, pricing inquiries, promotions, quotes.

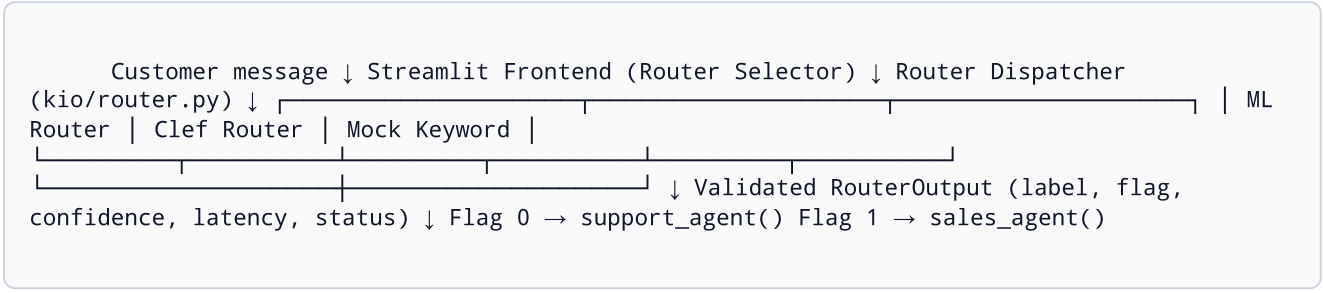
> **Mixed-Intent Rule**
> If a message contains both an existing problem and a purchasing interest, **Support takes priority** (routed to Flag 0).

The project compares two distinct approaches to making this single routing decision:

Approach	How It Decides
Decision AI (Cloudflare Clef)	Follows written routing instructions via zero-shot/prompt inference; no dedicated training dataset required.
Traditional ML	TF-IDF feature extraction + Logistic Regression, trained on 40 labelled synthetic messages.

* Note: This is a routing experiment, not a complete chatbot. Support and Sales agents are simulated by functions that print their names. No LLM answers, RAG pipelines, or live customer-service integrations are included.

2. System Architecture



All routers return an identical output structure, ensuring the frontend, test suites, and evaluation scripts handle them interchangeably.

3. Dataset Specification

Dataset location: `data/Agent_KIO_Dataset_FROZEN.xlsx`. Contains 60 synthetic English messages across 7 columns (ID, Message, Category, Flag, Split, Group_ID, Type).

Split	Total Messages	Support (0)	Sales (1)
Training	40	20	20
Test	20	10	10
Total	60	30	30

- Includes paraphrases, negation, and mixed-intent customer queries.
- Near-duplicates share a `Group_ID` and are strictly assigned to the same split to prevent data leakage.
- The data loader enforces validation checks on row counts, class balance, flag matches, group integrity, and a SHA-256 hash. Both models are evaluated on the **identical 20 test messages**.

4. Approach A: Decision AI (Clef)

- **Provider:** Cloudflare Workers AI using the Clef model (hosted solution, no fine-tuning performed).
- **Inputs:** Routing instructions (`kio/routing_instructions.md`), user message, and a binary choice prompt (Support vs Sales).
- **Output:** Selected choice, calculated probabilities, and confidence score.
- **Output Validation:** Strictly accepts only `support` or `sales`. Authentication errors, rate limits, timeouts, or malformed JSON result in an explicit *invalid* status—never a guessed fallback label.
- **License & Terms:** **[FILL IN from Cloudflare model page]**
- Instructions were finalized **[CONFIRM: before]** running the 20 test messages.

5. Approach B: Traditional Machine Learning

- **Pipeline:** TF-IDF Vectorizer (unigrams + bigrams, sublinear TF) → Logistic Regression Classifier.
- **Regularization:** Parameter C was optimized via 5-fold grouped cross-validation on training data only. Final saved model uses `C = 0.1`.
- **Performance:** Cross-validated training accuracy reached 82.5% (training set accuracy 100%). Removing an early 2D reduction step increased pipeline accuracy from 60% to 90%.
- **Artifact:** The saved model `models/ml_model.joblib` is the authoritative pipeline and was kept frozen during system integration.

6. Frontend Interface

Built with Streamlit (`frontend/app.py`) featuring a custom light purple theme. Key features include dynamic router selection, message example presets, confidence visualizers, latency badges, and a side-by-side **Compare Both** diagnostic view that flags disagreements between routers.

7. Comparative Test Results (20 Test Messages)

Metric / Feature	Clef (Decision AI)	Traditional ML
Correct / Incorrect	**20 / 0**	**18 / 2**
Accuracy	**100%**	**90%**
Confusion Matrix	[[10, 0], [0, 10]]	[[10, 0], [2, 8]]
Sales Precision / Recall / F1	1.00 / 1.00 / 1.000	1.00 / 0.80 / 0.889
Average Latency	~1,467 ms	**~8.8 ms** (~170x faster)
Typical Confidence	0.90 – 0.99	0.50 – 0.52
Infrastructure Requirement	Internet & Cloudflare API	Local Python runtime (Offline)

> **Disagreement Analysis (IDs 59 & 60)**
> The ML model failed on negation-heavy Sales requests (ID 59: *"I am not interested in TV, just give me the internet prices"*; ID 60: *"No TV channels, just pure fiber broadband, how much?"*), misrouting them to Support due to negative keywords. Clef correctly interpreted the intent.

8. Limitations & Recommendations

- **Data Volume:** With only 20 test messages, each error represents a 5% accuracy shift. 100% accuracy does not imply perfect real-world performance.
- **Recommendation:** Clef is superior for nuanced wording and negation. The ML model provides ultra-fast, zero-cost offline routing. A **hybrid architecture** is recommended: run the fast local ML model first, falling back to Clef whenever ML confidence is low or negation is detected. **[TEAM DECISION: Finalize wording]**

9. Team Responsibilities & Repository

- **Student 1:** Problem definition, architecture, dataset creation, labels, and split balance.
- **Student 2:** Decision AI (Clef) integration, routing instructions, and Decision API evaluation.
- **Student 3:** Traditional ML model training, feature pipeline, evaluation, and comparative analysis.